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AWS VPC Peering — Private Connectivity Between Two VPCs

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AWS VPC Peering | EWB Cloud Lab Project

A hands-on AWS networking project demonstrating private communication between two Amazon Virtual Private Clouds (VPCs) using VPC Peering, route tables, security groups, and EC2 connectivity testing.

Project Overview

This project implements VPC Peering between two VPCs in the AWS `eu-north-1` (Europe – Stockholm) region. The objective is to enable private communication between App-A and App-B using their private IP addresses, without assigning public IPv4 addresses to the application instances.

Project Objectives

- Create and configure two separate VPCs.
- Configure public and private subnets according to the project design.
- Deploy a Bastion Host and two application EC2 instances.

- Create and activate a VPC Peering connection.
- Configure routes in the relevant route tables.
- Configure security group rules for SSH and ICMP traffic.
- Test connectivity before and after peering, including controlled failure and recovery.
- Document the architecture, implementation, verification, and cleanup evidence.

AWS Services and Components

- Amazon VPC
- Amazon EC2
- VPC Peering
- Subnets and route tables
- Internet Gateway
- Security Groups
- SSH, ICMP, and Linux `ping` utilities

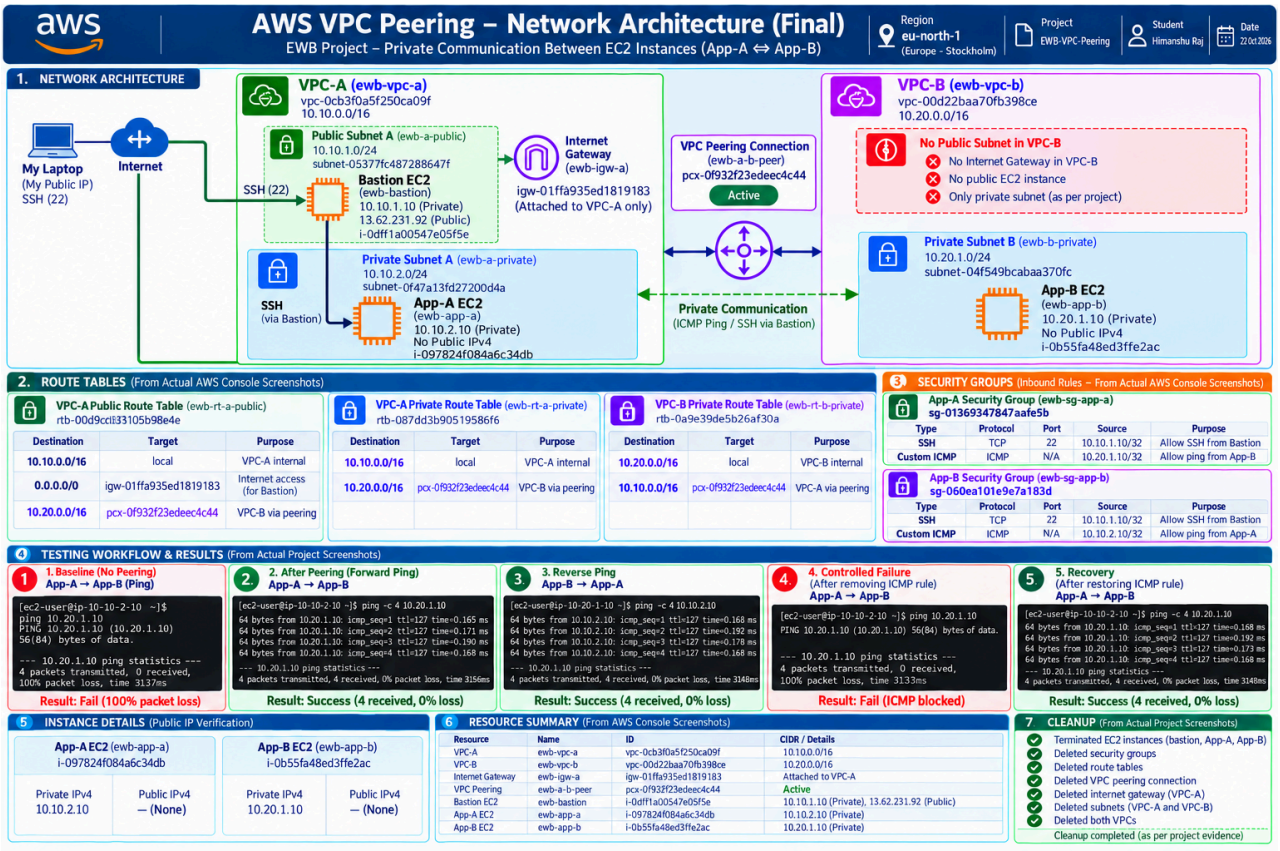
Network Configuration

Component	Configuration
AWS Region	`eu-north-1`
VPC-A	`ewb-vpc-a` — `10.10.0.0/16`
VPC-B	`ewb-vpc-b` — `10.20.0.0/16`
Public Subnet A	`10.10.1.0/24`
Private Subnet A	`10.10.2.0/24`
Private Subnet B	`10.20.1.0/24`
Bastion Host	`10.10.1.10` (private IP)
App-A	`10.10.2.10` (private IP)
App-B	`10.20.1.10` (private IP)
VPC Peering ID	`pcx-0f932f23edeec4c44`

Network design note: VPC-B does not include a public subnet or Internet Gateway in this project design. App-A and App-B communicate using private IP addresses through VPC Peering.

Architecture

The architecture connects VPC-A and VPC-B through a VPC Peering connection. Route tables provide routes to the remote VPC CIDR ranges. The Bastion Host provides the SSH jump-host path to the private instances, subject to the configured network and security group rules.



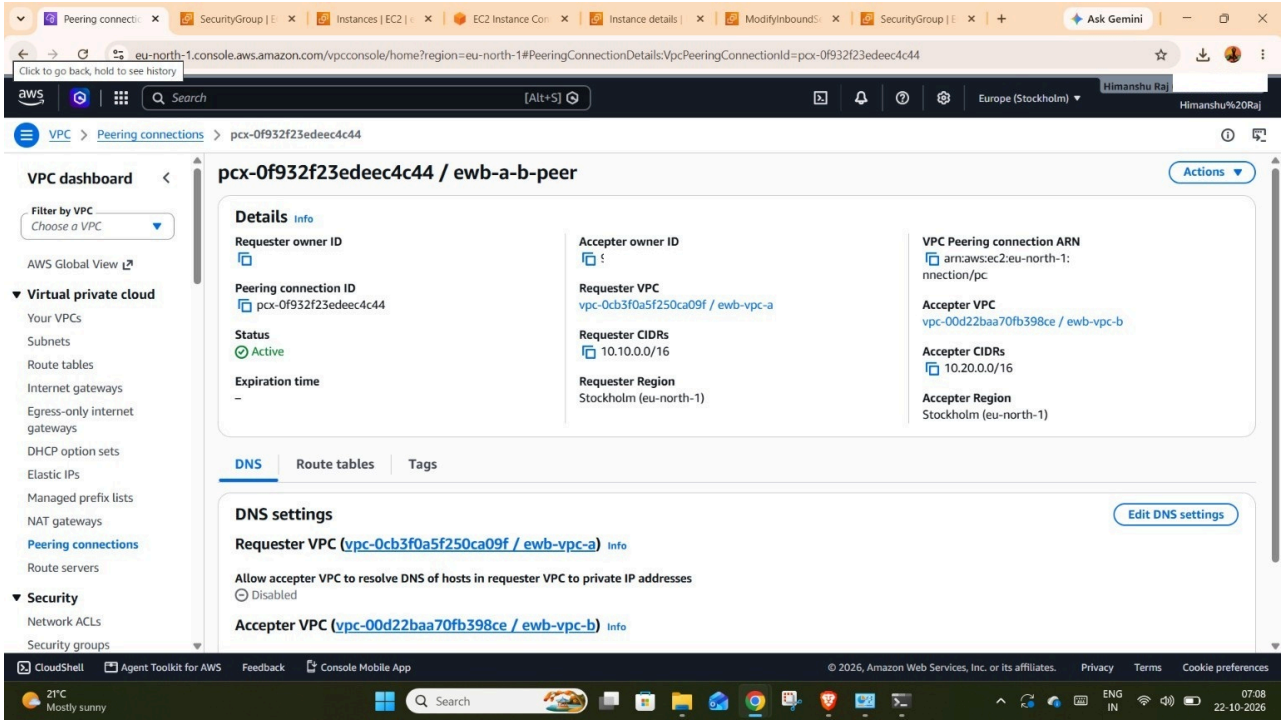
Implementation Steps

1. Created VPC-A and VPC-B in `eu-north-1`.
2. Configured the public subnet and private subnet in VPC-A, and the private subnet in VPC-B.
3. Launched the Bastion Host, App-A, and App-B EC2 instances.
4. Created and accepted the VPC Peering connection.
5. Added the required peering routes to the relevant route tables.
6. Configured SSH and ICMP security group rules.
7. Tested connectivity between App-A and App-B.
8. Removed the relevant ICMP rule for a controlled failure test, then restored the rule and tested recovery.
9. Reviewed resource configuration and documented cleanup evidence.

Verification and Testing

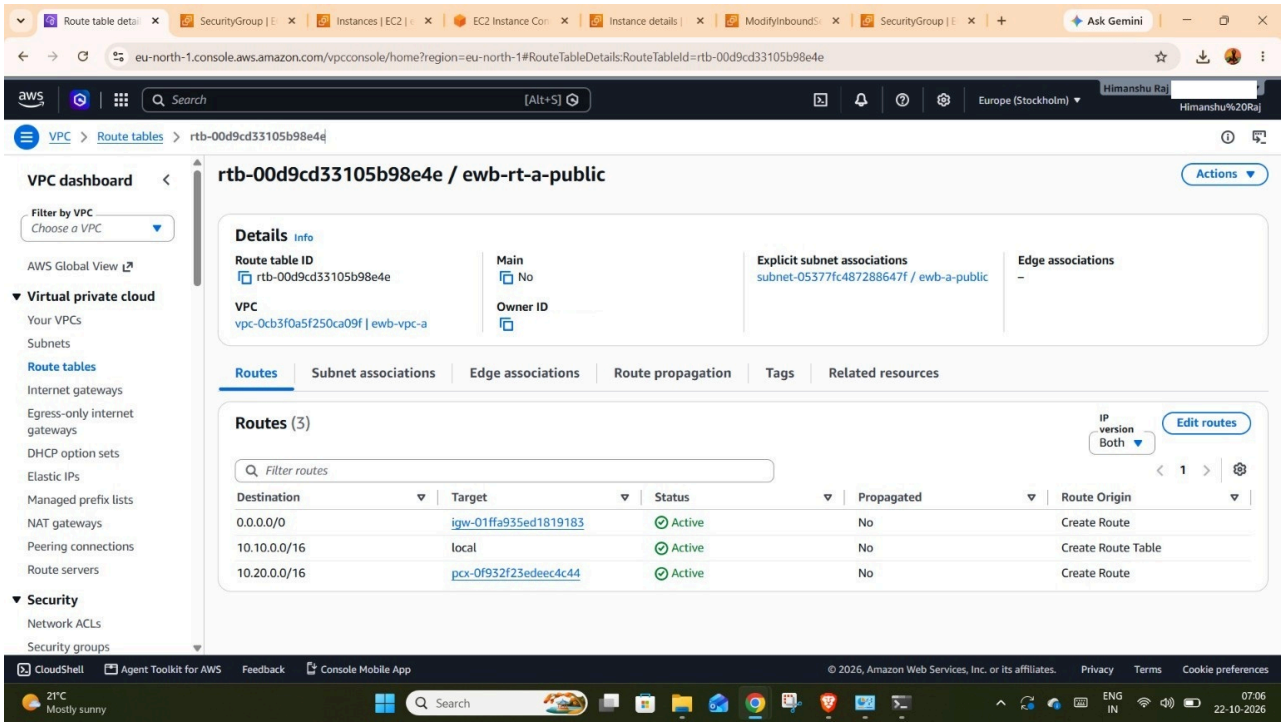
The following evidence highlights the key configuration and connectivity checks performed during the project.

1. VPC Peering Connection — Active

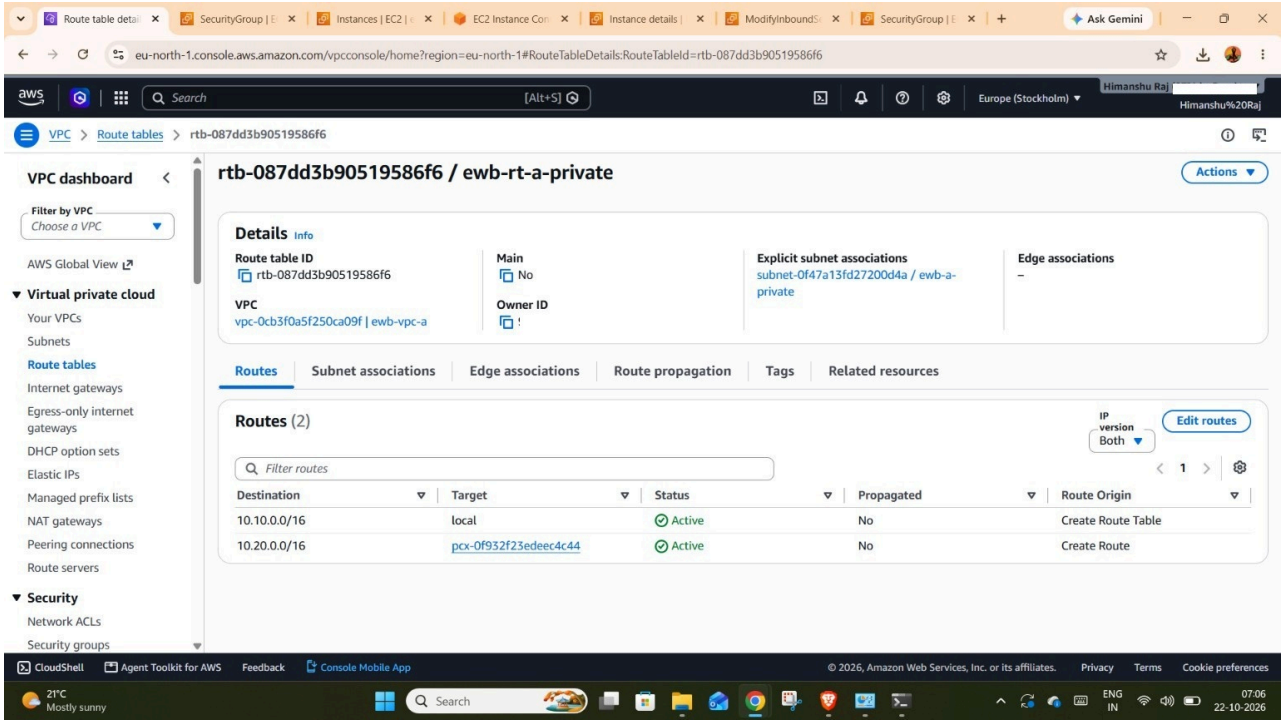


2. Route Table Configuration

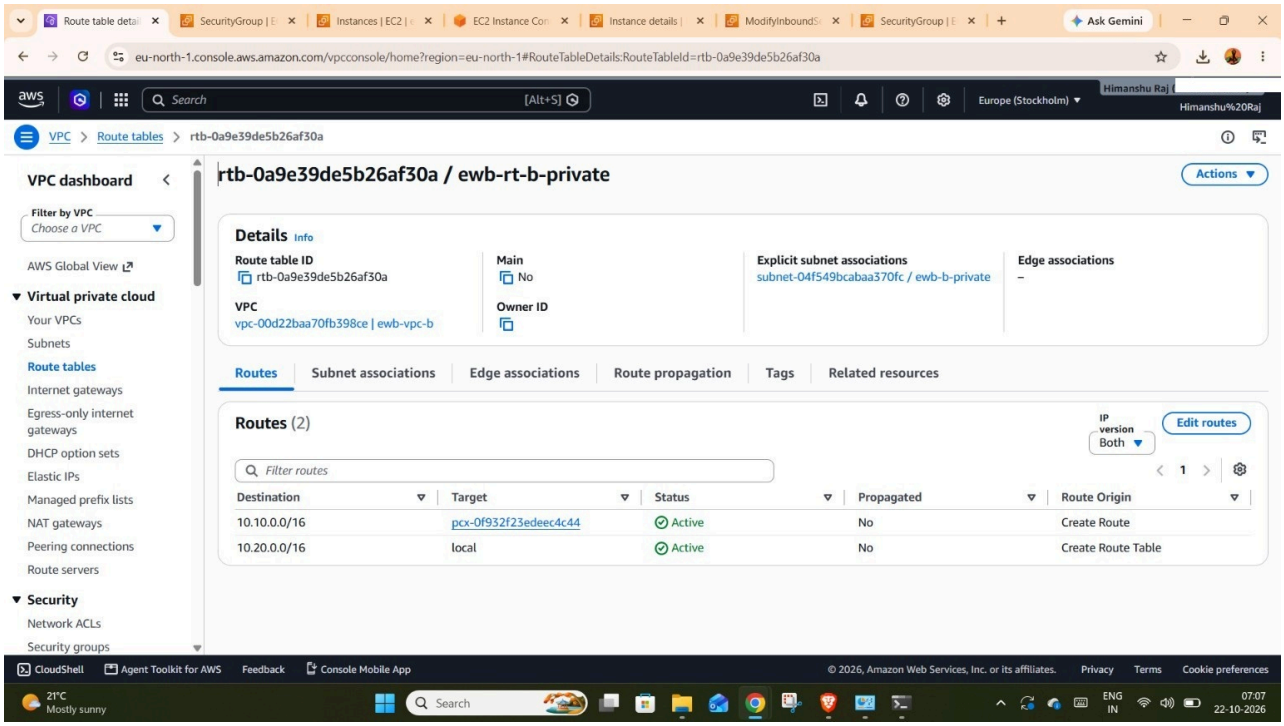
VPC-A Public Route Table



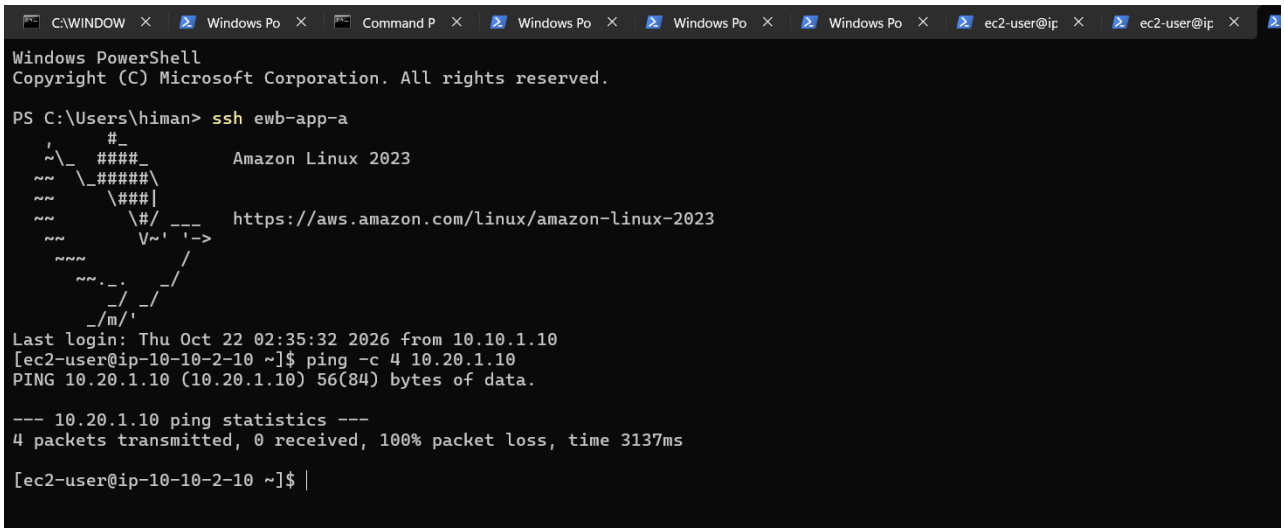
VPC-A Private Route Table



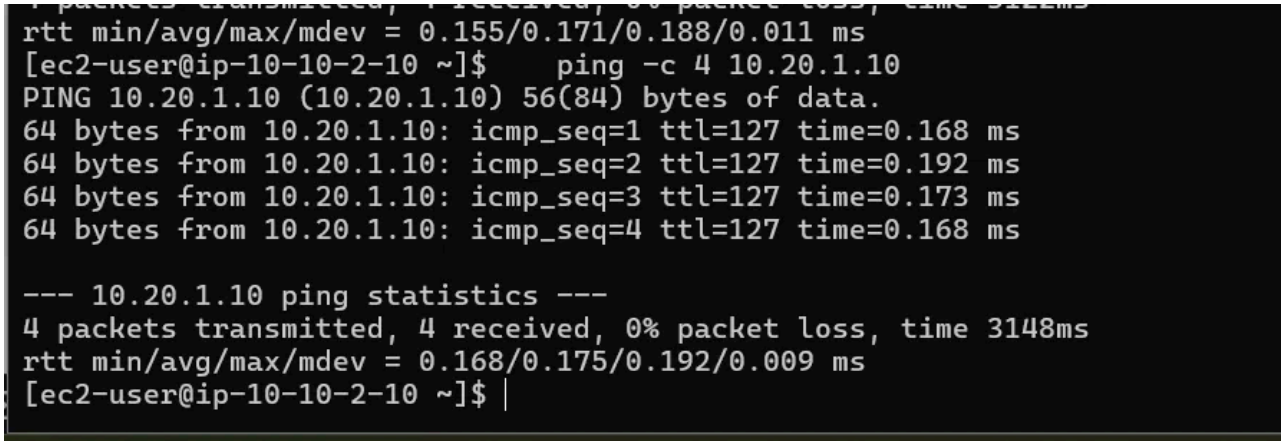
VPC-B Private Route Table



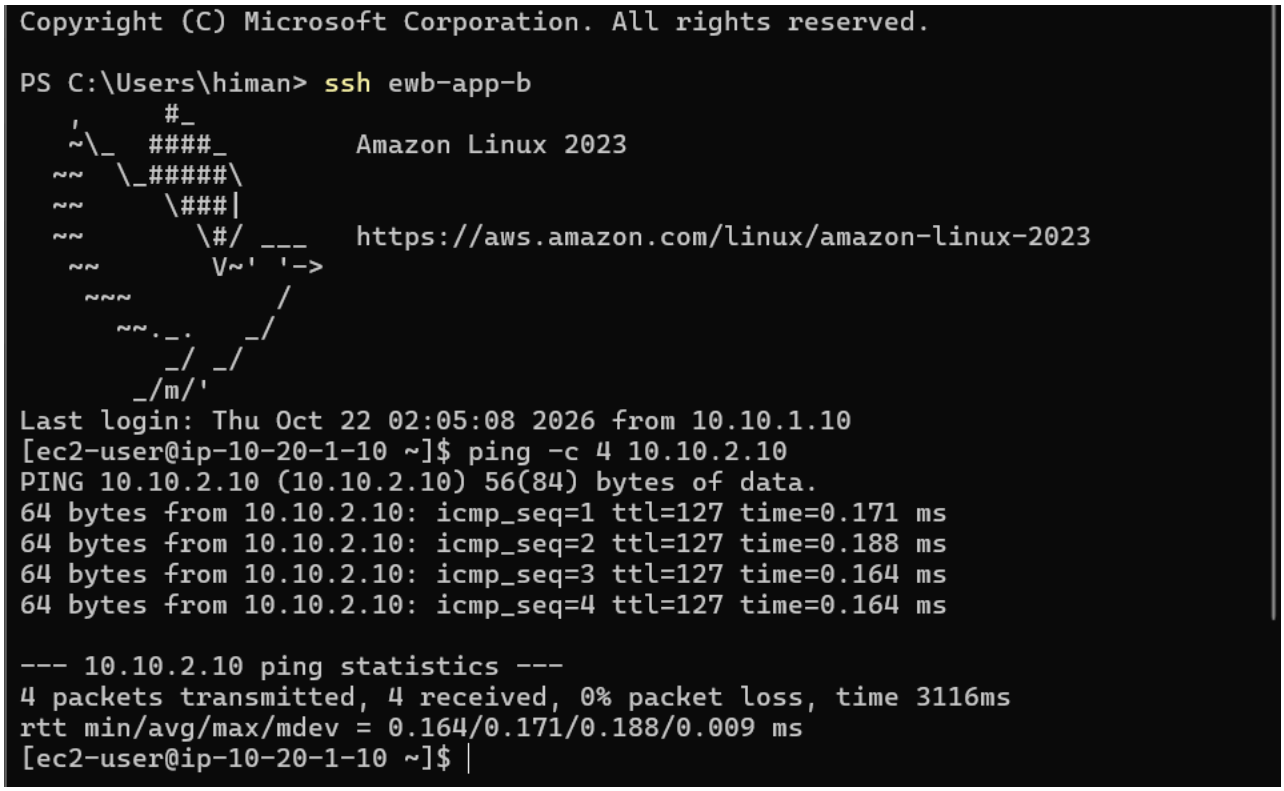
3. Initial Connectivity Failure



4. Successful Connectivity — App-A to App-B



README



6. Controlled Failure Test

```
rtt min/avg/max/mdev = 0.168/0.175/0.192/0.009 ms
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3133ms

[ec2-user@ip-10-10-2-10 ~]$
```

7. Connectivity Recovery

```
Last login: Thu Oct 22 00:59:08 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.165 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.171 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.190 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.168 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3156ms
rtt min/avg/max/mdev = 0.165/0.173/0.190/0.009 ms
[ec2-user@ip-10-10-2-10 ~]$
```

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Contributors 1



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